

## APPENDIX I

### PROPERTIES AND OPTIMIZATION INTERPRETATION OF FCPC

This appendix replaces the previous convergence argument with results that match the implemented mechanism. Local differential privacy (LDP) is applied only to the label histogram and sample-count metadata used for pairing; it is not applied to stochastic gradients. We therefore do not introduce an LDP gradient-variance term. Moreover, JSDN determines the matching and is not assumed to be linearly related to gradient dissimilarity. The relationship between JSDN and observed gradient divergence is evaluated empirically in the revised experiments.

#### A. Definition and Basic Properties of JSDN

Let  $\mathbf{p}_i, \mathbf{p}_j \in \Delta^{C-1}$  denote the label distributions of two clients, and let  $N_i, N_j > 0$  denote their sample counts. With logarithms in base two, define

$$\text{JSD}(\mathbf{p}_i, \mathbf{p}_j) = \frac{1}{2} D_{\text{KL}}(\mathbf{p}_i \| \mathbf{m}_{ij}) + \frac{1}{2} D_{\text{KL}}(\mathbf{p}_j \| \mathbf{m}_{ij}), \quad (1)$$

$$\mathbf{m}_{ij} = \frac{\mathbf{p}_i + \mathbf{p}_j}{2}, \quad (2)$$

$$q_{ij} = \frac{|N_i - N_j|}{N_i + N_j}, \quad (3)$$

$$\text{JSDN}_{ij} = (1 - \lambda) \text{JSD}(\mathbf{p}_i, \mathbf{p}_j) + \lambda q_{ij}, \quad \lambda \in [0, 1]. \quad (4)$$

*Proposition 1 (Boundedness and symmetry):* For all valid client pairs,  $\text{JSDN}_{ij} = \text{JSDN}_{ji}$  and  $0 \leq \text{JSDN}_{ij} \leq 1$ . Furthermore,  $\text{JSDN}_{ij} = 0$  if the two clients have identical label distributions and equal sample counts.

*Proof:* The Jensen–Shannon divergence with base-two logarithms is symmetric and lies in  $[0, 1]$ . The normalized quantity difference  $q_{ij}$  is also symmetric and lies in  $[0, 1]$  because  $|N_i - N_j| \leq N_i + N_j$ . JSDN is a convex combination of these two quantities, so it is symmetric and belongs to  $[0, 1]$ . If both metadata components are identical, both terms are zero. ■

*Proposition 2 (Single-skew reduction):* If  $N_i = N_j$  for all candidate pairs, ranking pairs by JSDN is identical to ranking them by JSD whenever  $\lambda < 1$ . If  $\mathbf{p}_i = \mathbf{p}_j$  for all candidate pairs, ranking pairs by JSDN is identical to ranking them by  $q_{ij}$  whenever  $\lambda > 0$ .

*Proof:* Under equal sample counts,  $q_{ij} = 0$  and  $\text{JSDN}_{ij} = (1 - \lambda) \text{JSD}_{ij}$ ; multiplication by the positive constant  $1 - \lambda$  preserves the ranking. Under equal label distributions,  $\text{JSD}_{ij} = 0$  and  $\text{JSDN}_{ij} = \lambda q_{ij}$ , for which the same argument applies. ■

These propositions justify the use of a fixed  $\lambda$  at the two single-skew limits, but they do not identify a universally optimal value in mixed-skew settings. Consequently, the revised manuscript treats  $\lambda = 0.3$  as an empirically selected default and reports sensitivity over heterogeneity levels and client scales instead of claiming a theoretical optimum.

#### B. Optimization of the Paired Local Objective

During a local update, client  $i$  treats the paired client's previous-round model  $\mathbf{r}_i^t = \mathbf{w}_{\pi_t(i)}^{t-1}$  as fixed and minimizes

$$\Phi_i^t(\mathbf{w}) = F_i(\mathbf{w}) + \beta \|\mathbf{w} - \mathbf{r}_i^t\|_2^2, \quad \beta \geq 0. \quad (5)$$

For the first communication round, the implementation defines the unavailable previous-round reference as the common initialization  $\mathbf{w}^0$ . All partner references are snapshotted before any client begins the current round, which prevents an order-dependent asymmetric update in sequential simulation.

*Proposition 3 (Local descent and stationarity):* Assume  $F_i$  is differentiable, bounded below, and  $L$ -smooth. Then  $\Phi_i^t$  is  $(L + 2\beta)$ -smooth. For the update

$$\mathbf{w}^{s+1} = \mathbf{w}^s - \eta \nabla \Phi_i^t(\mathbf{w}^s) \quad (6)$$

with  $0 < \eta \leq 1/(L + 2\beta)$ ,

$$\Phi_i^t(\mathbf{w}^{s+1}) \leq \Phi_i^t(\mathbf{w}^s) - \frac{\eta}{2} \|\nabla \Phi_i^t(\mathbf{w}^s)\|_2^2. \quad (7)$$

After  $S$  local steps,

$$\min_{0 \leq s \leq S} \|\nabla \Phi_i^t(\mathbf{w}^s)\|_2^2 \leq \frac{2 [\Phi_i^t(\mathbf{w}^0) - \inf_{\mathbf{w}} \Phi_i^t(\mathbf{w})]}{\eta S}. \quad (8)$$

*Proof:* The gradient of the quadratic term is  $2\beta(\mathbf{w} - \mathbf{r}_i^t)$  and is  $2\beta$ -Lipschitz. Hence  $\nabla \Phi_i^t$  is  $(L + 2\beta)$ -Lipschitz. Applying the descent lemma with  $L_\Phi = L + 2\beta$  gives

$$\Phi_i^t(\mathbf{w}^{s+1}) \leq \Phi_i^t(\mathbf{w}^s) - \eta \left(1 - \frac{L_\Phi \eta}{2}\right) \|\nabla \Phi_i^t(\mathbf{w}^s)\|_2^2 \quad (9)$$

$$\leq \Phi_i^t(\mathbf{w}^s) - \frac{\eta}{2} \|\nabla \Phi_i^t(\mathbf{w}^s)\|_2^2. \quad (10)$$

Summing over  $s = 0, \dots, S - 1$ , using the lower bound of  $\Phi_t^t$ , and upper-bounding the minimum by the average proves the final inequality. ■

This result establishes that the implemented quadratic paired objective has a standard nonconvex stationarity guarantee. It deliberately does not claim that larger JSDN causes a faster global convergence rate; that causal claim would require assumptions connecting metadata distributions to model gradients and is instead tested through controlled pairing ablations.

## APPENDIX II MATCHING GUARANTEE AND ODD-CLIENT HANDLING

For a participating client set  $V_t$ , construct the complete undirected graph  $G_t = (V_t, E_t)$  with nonnegative edge weight  $d_{ij} = \text{JSDN}_{ij}$ . FCPC greedily scans edges in nonincreasing weight order and accepts an edge if neither endpoint has already been matched.

*Theorem 1 (Greedy matching approximation):* Let  $M_g$  be the greedy dissimilarity matching and  $M^*$  a maximum-weight matching. For nonnegative JSDN weights,

$$\sum_{e \in M_g} d_e \geq \frac{1}{2} \sum_{e \in M^*} d_e. \quad (11)$$

*Proof:* Consider a greedy edge  $e = (i, j)$  when it is selected. At most two edges of  $M^*$  are incident to  $i$  or  $j$ . Any such optimal edge still available at this step has weight no larger than  $d_e$ , because the greedy scan is in nonincreasing order. Charge those at most two optimal edges to  $e$ . Every optimal edge is charged once, and the total charge to each greedy edge is at most  $2d_e$ . Thus  $w(M^*) \leq 2w(M_g)$ . ■

When  $|V_t|$  is odd, one client is necessarily unmatched by any ordinary matching. The revised FCPC procedure does *not* discard this client: it still performs local training and contributes to global aggregation, but does not receive the paired regularizer in that round. A fair-bye rule chooses a client with the smallest previous unmatched count before greedily matching the remaining clients. For an always-participating odd client set, the unmatched counts of any two clients therefore differ by at most one. This preserves the training participation rate and prevents one representative client from being systematically excluded from regularization.

## APPENDIX III COMPLEXITY AND COMMUNICATION ACCOUNTING

For  $n = |V_t|$  participating clients,  $C$  classes, and a model state containing  $P_b$  bytes:

- Building the dense JSDN matrix costs  $O(n^2C)$  arithmetic and  $O(n^2)$  storage. The implementation normalizes each histogram once and evaluates the matrix in bounded-memory vectorized blocks.
- Sorting the  $n(n-1)/2$  edges and scanning them for greedy matching costs  $O(n^2 \log n)$  time and  $O(n^2)$  storage. Exact maximum-weight matching is used only as an ablation because its worst-case cost is cubic.
- Computing the FCPC penalty is  $O(P)$  arithmetic over trainable parameters, where  $P$  is the parameter count, and requires one frozen partner state.
- Relative to FedAvg, FCPC does not add server communication rounds. However, a full-model implementation adds  $2|M_t|P_b$  partner-upload bytes per round, in addition to  $nP_b$  server downloads and  $nP_b$  server uploads. With an even participating set, the model-payload total is therefore  $3nP_b$  for FCPC versus  $2nP_b$  for FedAvg—a 50% increase in total model traffic, or one additional model upload per client. This cost is measured and reported rather than described as negligible.

The revised code logs JSDN construction and pairing time, round time, process CPU utilization, peak resident memory, GPU utilization, peak allocated GPU memory, per-link model bytes, and cumulative traffic. Scalability results for 10–1000 clients and accuracy–communication trade-offs are reported in the revised experimental section.

## APPENDIX IV THREAT MODEL

The main statistical-heterogeneity analysis assumes honest-but-curious clients and an authenticated server. LDP protects the uploaded metadata but does not make FCPC Byzantine robust. In particular, a malicious partner could supply a poisoned reference model. Recent ScaleSign and MSGuard results show that model-poisoning attacks and multi-strategy defenses require update-level robustness mechanisms [1]. Because FCPC does not modify server aggregation, it can be combined with robust aggregation or detection, but such a composition must filter both server-bound and partner-bound model states. The revised manuscript states this limitation explicitly and does not claim adversarial robustness from dissimilarity-aware pairing alone.

## REFERENCES

- [1] L. Yang, Y. Miao, Z. Liu, Z. Liu, X. Li, D. Kuang, H. Li, and R. H. Deng, “Enhanced model poisoning attack and multi-strategy defense in federated learning,” *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 3877–3892, 2025.